

Lab 1 – GradPace Product Description

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1 Introduction

Workforce expectations are changing rapidly due to automation, artificial intelligence, and global competition. Employers are no longer looking for just a degree. They expect candidates to have career-ready skills and relevant experience. According to the World Economic Forum, 63% of employers identify skill gaps as the biggest barrier to growth between 2025 and 2030 (World Economic Forum, 2025). At the same time, 66% of companies are decreasing entry-level hiring because of AI, and 91% report that automation is changing or eliminating jobs (Laurent, 2026). These shifts are already affecting graduates, with 50% to 60% underemployed and 48% feeling underqualified for entry-level roles in their field (Cengage Group, 2025). These figures highlight a growing disconnect between education and workforce expectations.

Although universities provide many resources, career preparation remains fragmented and is mainly student-driven. Academic progress is clearly structured, but professional development is not. Existing tools only address parts of the problem. Degree tracking systems focus on coursework, job boards list opportunities, and networking platforms showcase experience, but none of them guide students on how to connect these pieces into a clear career path. As a result, students identify gaps too late and make reactive decisions.

The main issue is the lack of a unified system that actually connects academic progress, professional development, and industry expectations. Students need a platform that provides continuous guidance, identifies gaps early, and offers personalized recommendations. GradPace addresses this need by serving as a unified career readiness platform that is designed to bridge academic progress and professional development, allowing students to identify skill gaps early and take guided steps towards resolving them.

2 Product Description

GradPace is a web-based career readiness platform designed to help college students connect their academic progress with the skills and experience expected by the modern workforce. Unlike normal academic tracking systems and job boards, GradPace focuses on proactive career preparation by evaluating a student's current academic and professional background. It then identifies steps they can take to become more competitive for their desired career. Students provide information such as their academic history, skills, career interests, and location, and then GradPace uses that to generate a personalized career roadmap for them. The platform also uses current industry data to compare a student's qualifications against real-world expectations and identify potential gaps before they enter the job market. The goal of GradPace is to give students a centralized and continuously updated view of their career readiness. This allows them to make informed decisions about coursework, skill development, professional opportunities, and networking.

2.1 Key Product Features and Capabilities

GradPace provides several features that work together to create a personalized career preparation experience. Users can create and manage an account and enter their academic and professional information. This includes their coursework, skills, experience, career interests, and location. Based on this information, the system generates a personalized roadmap that provides recommended steps toward the user's career goals. Users can then view this roadmap through an interactive dashboard and track their progress over time.

One of GradPace's most significant capabilities is its early-warning gap detection system. Instead of waiting for a student to begin applying to jobs to discover that they are missing

important skills or qualifications, GradPace identifies potential gaps earlier in the student's academic career. The platform can also provide an industry readiness dashboard that compares the user's current qualifications with standards and expectations found in the workforce. This allows students to understand not only what they have already done, but also what they should work toward next.

GradPace also makes use of live industry data and location-based recommendations. Industry professionals and alumni can provide information about current workforce trends, including skills and qualifications that employers are looking for. Alumni can participate in surveys about these trends, which helps the platform stay relevant as industry expectations change. Location-based services can also identify professional events and networking opportunities that may be available near the user. Together, these features allow GradPace to provide recommendations based on current workforce needs instead of relying on static academic requirements or general career advice.

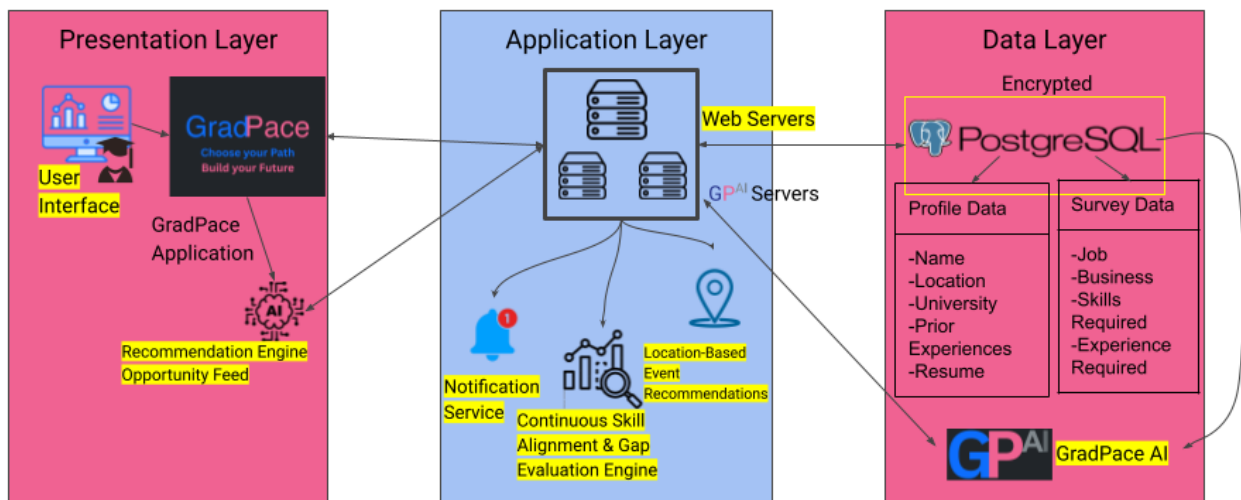
The primary goal of these features is to address the fragmented nature of existing career development tools. Coursework systems mainly track academic progress, job boards focus on available positions, and networking platforms focus on building professional connections. GradPace combines these areas into a single system centered around career readiness. By identifying gaps early, providing personalized recommendations, and using current industry information, the platform allows students to take a proactive approach to career development. This helps them avoid situations where they are reacting to employment requirements after graduation.

2.2 Major Components (Hardware/Software)

GradPace is designed as a cloud-based web application, which means users do not need specialized hardware to access the system. A standard computer or mobile device with an internet connection and web browser can access the platform. Most of the application's processing, storage, and external data services are hosted in the cloud. This method allows users to access their career information and roadmap without requiring more software or hardware.

The system is built around four main software components: the frontend, backend, database, and API integrations. The frontend provides the user interface through which users can create accounts, enter information, view their roadmap, monitor their career readiness, and get recommendations. The backend has the main application logic and manages functions such as authentication, security, roadmap generation, gap detection, industry-readiness comparisons, and development recommendations.

The database provides storage for the information required by the platform. GradPace will use PostgreSQL to store information such as user profiles, academic records, professional experience, skills, career goals, roadmap progress, and relevant opportunity information. This allows the backend to retrieve and update user information as the student progresses through the platform.

Figure 1*GradPace Major Functional Component Diagram*

The final major component consists of API integrations and external data services.

GradPace will connect to external services to obtain current industry information, location-based opportunities, and other data needed to support recommendations. AI-based systems may also be added into the platform to help with roadmap generation and refinement. These components work together through the system architecture shown in **Error! Reference source not found.**, with the frontend communicating with the backend, the backend managing application logic and database operations, and the external APIs giving additional information as needed.

3 Identification of Case Study

GradPace is being developed primarily for college students who want to prepare for their careers but are uncertain about what steps they should take beyond completing their coursework. The product is particularly intended for students who are early in their academic careers and may not yet have significant professional experience or a clear understanding of the skills and qualifications expected in their desired industry. By connecting academic information, career

interests, professional development, and current industry expectations, GradPace is intended to help these students identify opportunities and potential skill gaps early enough to address them before graduation.

Table 1

Table of Comparison Between RWP and Prototype

Features	Prototype	Real World Product
Login/Authentication	Full: Core functionality and easy to implement	Secure authentication system
Account Creation/Deletion	Full: Required for all users	Full account lifecycle management
Account Subscription Management	Partial: Simulated or mocked payments	Paid subscription system
Input Academic and Professional Info	Full: Core input for system logic	User profile with professional + academic data
Personalized Roadmap Generation	Partial: Simplified logic instead of full AI	Career roadmap
View Roadmap	Full: UI-based and feasible	Interactive roadmap UI
Early-Warning Gap Detection	Partial: Rule-based instead of AI-driven	Skill gap analysis
Industry Readiness Dashboard	Partial: Basic scoring instead of dynamic benchmarking	Visual readiness scoring system
Recommendations/Guided Next Steps	Full: Static or rule-based suggestions	Guided recommendations
Notifications/Alerts	Partial: Basic notifications only	Alerts and reminders
Location-Based Event Recommendations	Partial: Static or mock event data and smaller AI model for recommendations	GPS + scraped local opportunities
Input Up-To-Date Industry Data	Eliminated: Not feasible for prototype	Continuous industry data input system
Alumni Survey	Partial: Simple form implementation	Post-graduation feedback system
Survey Results Dashboard	Partial: Basic visualization only	Aggregated alumni feedback dashboard
User Management	Partial: Limited admin interface	Admin control panel
AI Configuration/Rule Updates	Eliminated: Too complex for prototype	Dynamic AI tuning system
Data Quality Monitoring	Partial: Manual or basic validation	Automated data validation system

The initial case study group will consist of a small group of undergraduate students who will use the GradPace prototype and provide feedback on its functionality and usefulness. These students will represent the intended student user population and will test the features listed in Table 1, including academic and professional information entry, personalized roadmap generation, opportunity recommendations, and career-readiness information. Their feedback will be used to evaluate whether the prototype is useful and effective at helping students know what actions to take toward their career goals. The case study will focus on students who may have different levels of professional experience and different career goals. Recruiting students with different levels of experience will help the team identify common usability and career-planning needs.

A typical student might begin by entering their academic background, skills, career interests, and location into GradPace. The platform then generates a personalized roadmap and identifies potential skill or experience gaps. Throughout the semester, the student can return to the platform to track progress, find relevant opportunities or networking events, and update their information as they gain new experience. GradPace uses these updates and current industry data to refine its recommendations, giving the student an ongoing view of what they can do next to prepare for their career.

In the future, GradPace could be expanded to support additional groups beyond undergraduate students. University career services departments and academic advisors could use the platform to help monitor student career readiness and provide more specific guidance. Alumni could participate by providing information about industry trends and the skills they have found valuable in their careers. Industry professionals could also contribute current information about workforce expectations and qualifications. Universities could eventually integrate

GradPace with existing academic systems, allowing the platform to combine academic progress with career preparation at a larger scale. These new users would allow GradPace to develop from a student-focused career planning tool into a wider system that could connect students, universities, alumni, and industries.

4 Glossary

API (Application Programming Interface): A set of rules that allow different software applications to communicate and exchange data with each other.

Authentication: The process of verifying a user's identity before allowing access to a system.

Backend: The server-side part of a website or application that handles logic, databases, and data processing.

Career Readiness: A measure of how prepared a student is to enter a specific career based on skills, experience, and qualifications.

Career Roadmap: A structured, visual plan that outlines a student's progress, goals, and required steps toward a specific career path.

Cloud Computing: The use of remote servers on the internet to store data and run applications instead of a local computer.

Data Leak: The accidental or unauthorized exposure of sensitive or private information.

Database: An organized collection of data that can be stored, accessed, and managed electronically.

Encryption: A security method that converts data into a coded form so that only authorized parties can read it.

Frontend: The user-facing part of a website or application that users interact with in their browser.

Gap Detection (Early-Warning System): A feature that identifies missing skills, experiences, or qualifications needed for a target career.

Industry Readiness Benchmarking: The process of comparing a student's profile against

expected skills and qualifications for specific careers.

Location-Based Event Recommendations: Suggestions for nearby internships, networking events, or career opportunities based on user location.

Professional Data Input: User-provided information such as coursework, projects, internships, and certifications used to build a profile.

Survey Results Dashboard: A feature that displays aggregated insights from alumni data to inform students and administrators.

System Administrator: A role responsible for managing the platform, maintaining data quality, and overseeing system functionality.

Token (AI): A small unit of text processed by an AI model, such as a word, part of a word, or punctuation.

User Interface (UI): The visual elements of a software application that allows users to interact with it.

Web Server: A computer system or software that stores websites and delivers web pages to users when they request them through a browser.

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